

CSE0611215 : Digital Logic Design

Part A: Course Information

1. **Program:** BSc. in CSE
2. **Course Code:** CSE0611215
3. **Course Title:** Digital Logic Design
4. **Course Type (GED/Core Course/Electives):** Core Course
5. **Year/Level/Semester/Term:** 1st year/ 3rd Semester
6. **Academic Session:** Autumn 2026
7. **Section:** 3A+3B+3C
8. **Prerequisite Course (If any):** Discrete Mathematics
9. **Credit Value:** 3.00
10. **Contact Hours:** 3.00 hours/Week
11. **Class Schedule:** Tuesday (Room: 203, Period: 10.00 AM – 12.00 PM),
Thursday (Room: 509, Period: 10.00 AM -11.00 AM)
12. **Counseling Schedule:**
 - i. Sunday: 8.00AM-10.00PM
 - ii. Monday: 11.00AM-1.00PM
 - iii. Wednesday: 9:00AM -10:00AM

1. **Course Teacher:** Farah Ulfat Zaima, Designation:Lecturer, Email: farah.ulfat@uits.ac.bd, Cell: +8801977799407, Specialization: Machine Learning, Deep Learning, Computer Vision, Image Processing.

Rationale of the Course: This course introduces a different numbering system, Boolean function, K-map and minimization techniques. This course also introduces different combinational and sequential circuits. Students will learn to design decoder, encoder, multiplexers, memory elements etc.

13. Course Objectives:

- a. To study the basic philosophy underlying the various number systems, negative number representation, binary arithmetic, binary codes and error detecting and correcting binary codes.
- b. To study representation of switching functions using Boolean algebra, expressions and their minimization techniques.
- c. To study the combinational logic design of various logic and switching devices and their realization.
- d. To study the sequential logic circuits design both in synchronous and Asynchronous modes.

Part B: Skill Mapping

14. Course Outcome (COs)

No.	COs	BT	CP	CA	KP	AA
CO1	Explain the fundamentals of numbering systems and Boolean algebra and simplification..	L2	-	-	1,2,3	CT, MT
CO2	Analyze the operation of different types of combinational and sequential logic circuits.	L4	-	-	3,4	MT, FE, A
CO3	Design different combinational and sequential logic circuits by determining their state behaviour and	L6	1,2,3	-	3,5,6	MT, FE, A

	justifying the appropriate configuration for a specified requirement.					
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*COs- Course Outcome; BT- Bloom's Taxonomy; CP/WP- Complex Problems; EA/WA- Complex Activities; AT- Assessment Tools; KP/WK- Knowledge Profile; DM&A: Delivery Methods & Activities; FE- Final Exam; CT- Class Test, MT- Mid Term, A- Assignment.

15. Mapping of COs with Program Outcomes (POs)

Course Outcomes (CO)	PO(a)	PO(b)	PO(c)	PO(d)	PO(e)	PO(f)	PO(g)	PO(h)	PO(i)	PO(j)	PO(k)	PO(l)
CO1	√											
CO2		√										
CO3			√									

Part C: Teaching Learning Approach

16. Course plan specifying contents, COs, co-curricular activities (if any), teaching learning and assessment strategy mapped with COs.

Week	Topic	Teaching-Learning Strategy	Assessment Strategy	Corresponding CLOs
1	Introduction to Digital System and number system. Number conversion and 1's, 2's complement addition subtraction.	Introduce digital systems and number systems with real-life examples, followed by number conversion exercises.	Midterm	CO1
2	Boolean algebra, De Morgan's theorem, Logic gates and truth table	Explain Boolean algebra, logic gates, and De Morgan's theorem using truth tables and worked examples.	Midterm	CO1
3	Standard and Canonical forms of Boolean functions, SOP & POS	Teach SOP, POS, standard and canonical forms through step-by-step derivation and problem-solving.	Midterm, Class Test 1	CO1
4	K- Map for two, three and four variables, don't care condition	Demonstrate K-map simplification for 2-4 variables and solve examples with don't-care conditions.	Midterm, Class Test 1	CO1
5	Design Procedure of Combinational Circuit, Half to Full Adders, Half to	Explain combinational circuit	Midterm	CO2, CO3

	Full Subtractors	design procedure and demonstrate the design of adders and subtractors.		
6	Coder Conversion, Multilevel NAND and NOR circuit design	Introduce code conversion techniques and implement Boolean functions using NAND and NOR gates.	Final Exam	CO1
7	Binary Parallel Adder cum Subtractor, De-Multiplexer, Encoder, Decoder, Multiplexer. Adder cum Subtractor.	Explain binary subtraction methods and design adder/subtractor, encoder, and de-multiplexer circuits.	Final Exam	CO2, CO3
8	Priority Encoder, Boolean function implementation with multiplexers,	Demonstrate the operation and design of priority encoders, multiplexers, and Boolean function implementation using multiplexers.	Final Exam	CO2, CO3
9	Comparator	Explain the operation and applications of comparators.	Final Exam	CO2, CO3
10	Basic Flip Flop Circuit, S-R flip flop, D Flip Flop, J-K Flip Flop, T flip flop	Introduce flip-flops and explain their characteristics, timing diagrams, and practical applications.	Final Exam, Assignment	CO2, CO3
11	Introduction to register, shift register, serial transfer	Explain the operation and applications of registers, shift registers, and serial data transfer with examples.	Final Exam, Assignment	CO2, CO3
12	Basics of Counter, Ripple Counter, Synchronous and Asynchronous Counter (Up Counter, Down Counter, BCD Counter), State diagram and table	Explain the design and operation of ripple, synchronous, asynchronous, and BCD counters using state diagrams and timing analysis.	Final Exam, Assignment	CO2, CO3
13	Revision of the whole syllabus	Revision and QNA	Final Exam	CO1, CO2, CO3

Part D: Assessment Approach

17. Assessment and Evaluation

Assessment components		Marks Distribution	COs	Bloom's Taxonomy
Continuous Assessment (50%)	Class Attendance/Class Performance	10%	-	-
	Class Test/Assignment/Quiz/Surprise Test	20%	-	-
	Mid Term	20%	-	-
Comprehensive Final Exam		50%	-	-
			-	-
			-	-
Total Marks		100%		

18. Assessment Pattern

1. Continuous Internal Evaluation (50 Marks)

*The percentage distribution of Bloom's categories in the assessment tools may vary by $\pm 5\%$.

Bloom's Category	Class Test 1/ Assignments 1	Mid Term Exam	Class Test 2/ Assignments	Attendance & Class Participation
Marks	(10)	(20)	(10)	(10)
Remember		2%		
Understand	5%	6%		
Apply	5%	8%	5%	
Analyze		4%	5%	
Evaluate				
Create			5%	

2. Final Exam Evaluation (50 Marks)

*The percentage distribution of Bloom's categories in the assessment tools may vary by $\pm 5\%$.

Bloom's Category	Final Term (50)
Remember	
Understand	10%
Apply	10%
Analyze	10%
Evaluate	5%
Create	15%

19. Grading system

Numerical Grade	Letter Grade	Grade Point
80% and above	A + (A plus)	4.00
75% to less than 80%	A (A regular)	3.75
70% to less than 75%	A – (A minus)	3.50
65% to less than 70%	B + (B plus)	3.25
60% to less than 65%	B (B regular)	3.00
55% to less than 60%	B – (B minus)	2.75
50% to less than 55%	C + (C plus)	2.50
45% to less than 50%	C (C regular)	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Part E: References

20. Learning Materials

- 1) **Recommended Readings:** Lecture notes and Necessary documents (Uploaded in Google Classroom)
- 2) **Supplementary Readings:** Some online links would be provided for further study,
Digital Logic and Computer Design- Morris Mano
- 3) **Others** (as applicable for the discipline academic program): N/A.

Part F: Additional Information

1. **Program Outcome:** <https://uits.ac.bd/public/cse>
2. **Outcome Based Curriculum:** https://uits.ac.bd/public/obe_curriculum_cse
3. **Course Catalog:** https://uits.ac.bd/public/course_catalog_cse
4. **Faculty Information:** <https://uits.ac.bd/public/faculty-members-cse>
5. **Assessments Rubrics:**
6. **Details of Bloom Taxonomy, Knowledge Profile, Complex Engineering Problems and Complex Engineering Activities:**
7. **Make-up Exam Procedure:**
8. **Attendance Policy:**
9. **Academic Calendar:** <https://uits.ac.bd/academic-calendar/>
10. **Academic Policies:**
11. **Proctorial Rules (Code of Conduct):** <https://uits.ac.bd/wp-content/uploads/2024/01/Code-of-Conduct.pdf>